Data Structures and Algorithms - 2025 - 2026

Quiz navigation

1 ⊗	2 ✓	3	4 ⊗	5
6 ✓	7 ✓	8 ✓	9 ✓	10 ✓
11 ✓	12 ⊗	13 ✓	14 ✓	15 ⊗
16 ✓	17 ✓	18 ✓	19 ⊗	20 ⊗
21 ⊗				

Show one page at a time

Finish review

Status	Finished
Started	Sunday, 17 May 2026, 3:10 PM
Completed	Sunday, 17 May 2026, 3:28 PM
Duration	17 mins 40 secs
Marks	10.77/21.00
Grade	**5.13** out of 10.00 (**51.27**%)

Question **1**
Incorrect
Mark 0.00 out of 1.00
Flag question

What is the result of evaluating the following postfix expression containing single digit numbers and the usual operators: 4 3 + 9 - 1 2 4 * + * 3 1 2 - * + ?

Answer: 44 ⊗

In seminar 5 we have talked about how to evaluate an arithmetic expression.

The correct answer is: -21

Question **2**
Correct
Mark 1.00 out of 1.00
Flag question

What is the postfix notation for the following expression (which is in infix notation): (1 * 3 + 4) - (2 * (6 + 2)) * 2 ?

- ○ 1 3 4 2 6 2 2 * + * + *
- ○ 1 3 * 4 + 2 * 6 2 + - 2 *
- ○ 1 3 * 4 + 2 6 * 2 + 2 * -
- ○ 1 3 * 4 + - 2 * 6 2 + 2 *
- ● 1 3 * 4 + 2 6 2 + * 2 * - ✓

Your answer is correct.

Algorithm for transformation was discussed in Seminar 5.

The correct answer is:
1 3 * 4 + 2 6 2 + * 2 * -

Question **3**
Partially correct
Mark 0.17 out of 1.00
Flag question

Which of the following arithmetic expressions in postfix notation are correct?

- ☑ 3 2 1 * 3 * - 9 2 * - ✓ It is correct. The result of evaluating it is -21.
- ☐ 2 3 4 + * + 3 2 -
- ☐ 3 2 1 * - 9 2 * 4
- ☑ 1 2 3 * 3 - ⊗ Not correct. At the end of the evaluation we will have two elements left in the stack, when we should have only one.
- ☐ 2 3 4 + * 3 + 2 -

Your answer is partially correct.

You have correctly selected 1.
The easiest way to see if an expression is correct or not is to try to evaluate it. If evaluation goes well, the expression is correct.

The correct answers are:
2 3 4 + * 3 + 2 -,

3 2 1 * 3 * - 9 2 * -

Question **4**
Incorrect
Mark 0.00 out of 1.00
Flag question

Assume that we transform the expression ((3 + 5 * 2) * (4 + 1 * 7)) to the postfix notation using the algorithm discussed at Seminar 5. What is the maximum number of items we will have on the stack at any point during the transformation?

- ● 4 ⊗
- ○ 1
- ○ more than 6
- ○ 2
- ○ 6
- ○ 3
- ○ 5

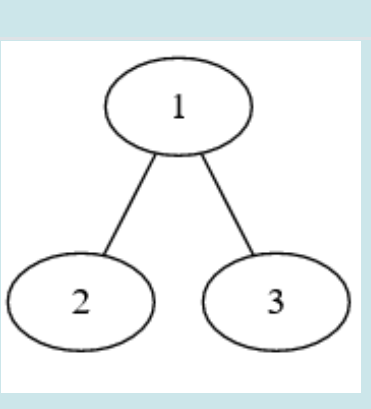
Your answer is incorrect.

Right before processing the) after 7, the content of the stack is: (* (+ * .

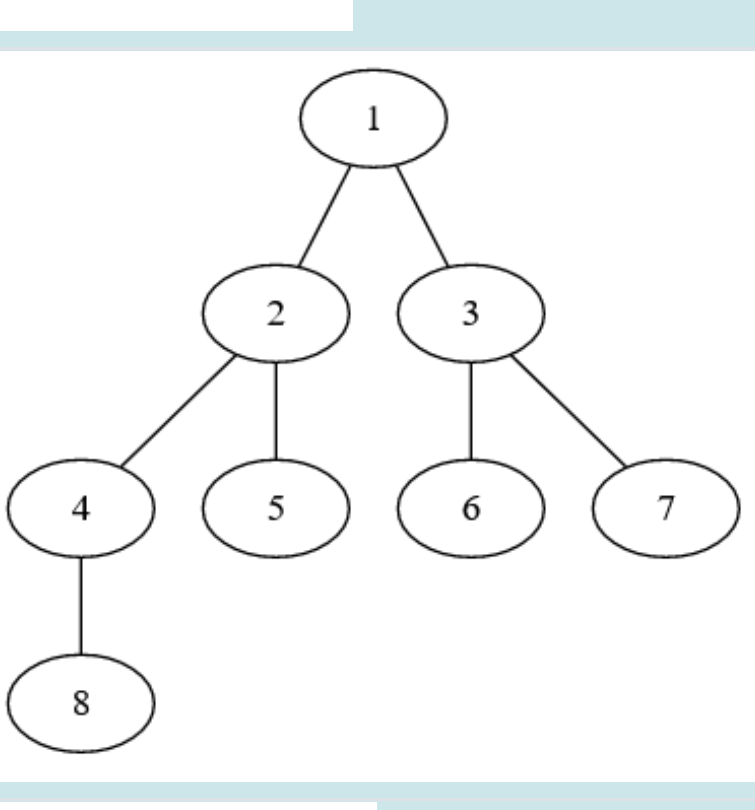
The correct answer is:
5

Question **5**
Partially correct
Mark 0.60 out of 1.00
Flag question

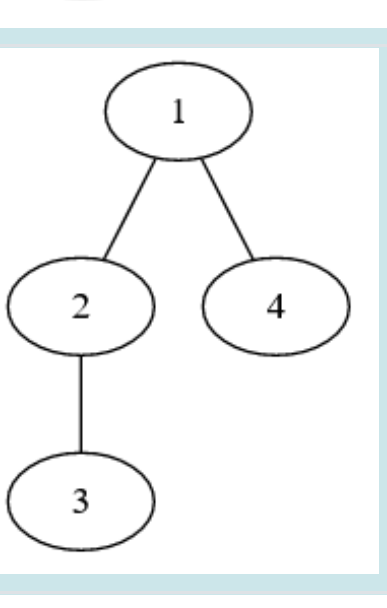
Match the figures with the corresponding description. Look only at the structure of the trees, ignore the values from the nodes.



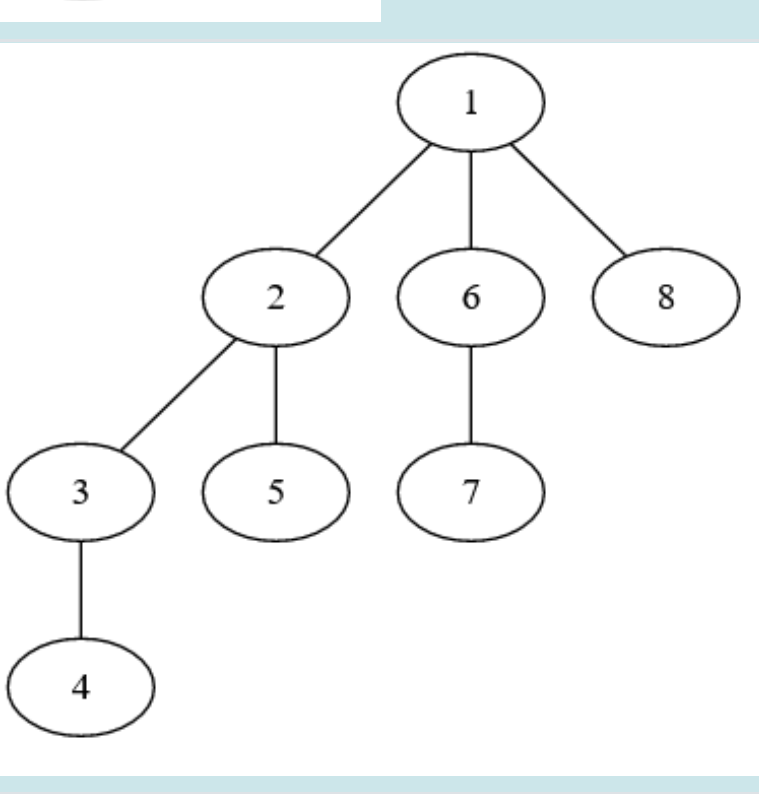
Binomial tree of order 2 ⊗



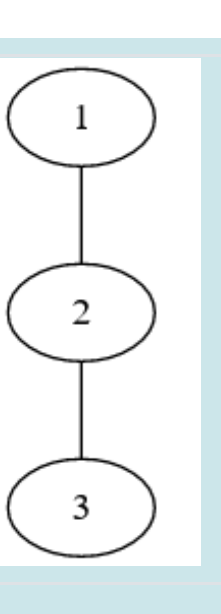
Binomial tree of order 3 ⊗



Binomial tree of order 2 ✓



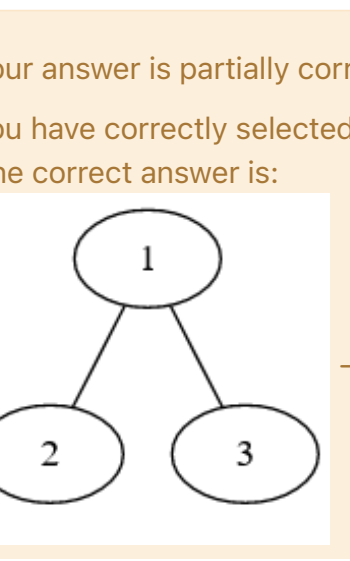
Binomial tree of order 3 ✓



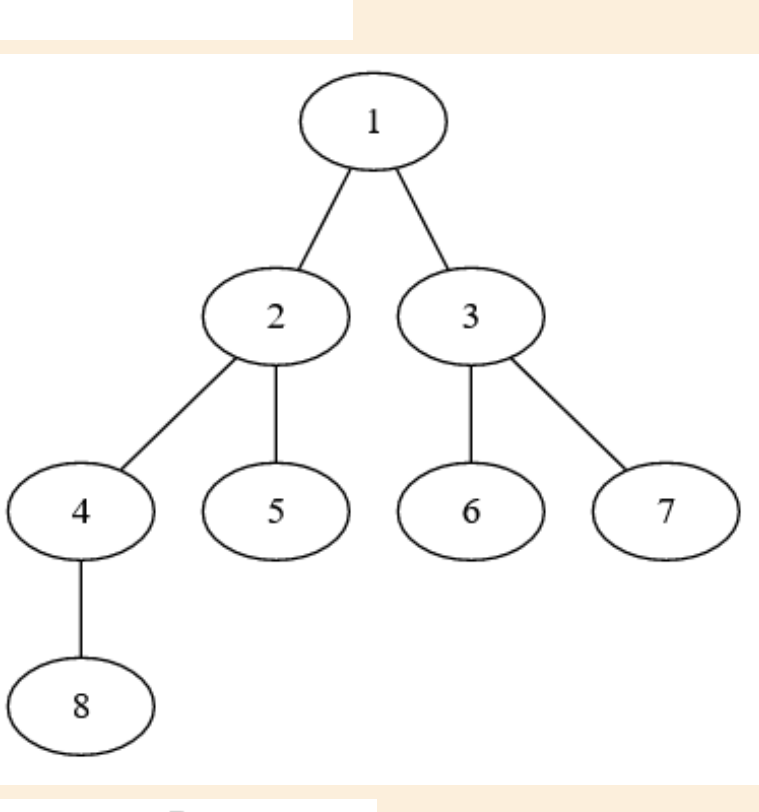
Neither binary heap nor binomial tree ✓

Your answer is partially correct.

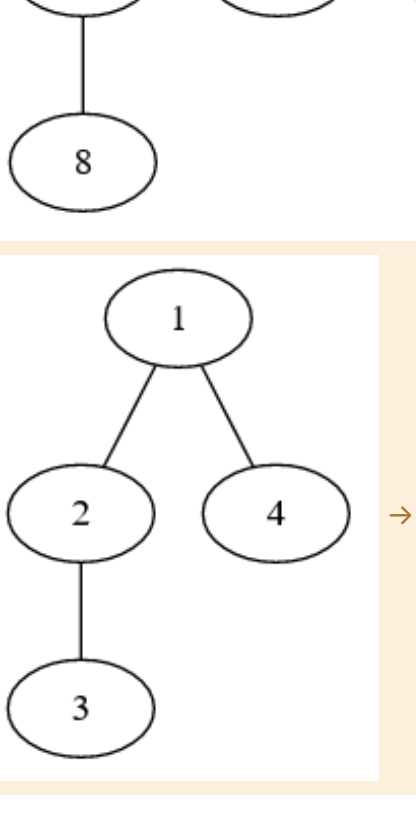
You have correctly selected 3.
The correct answer is:



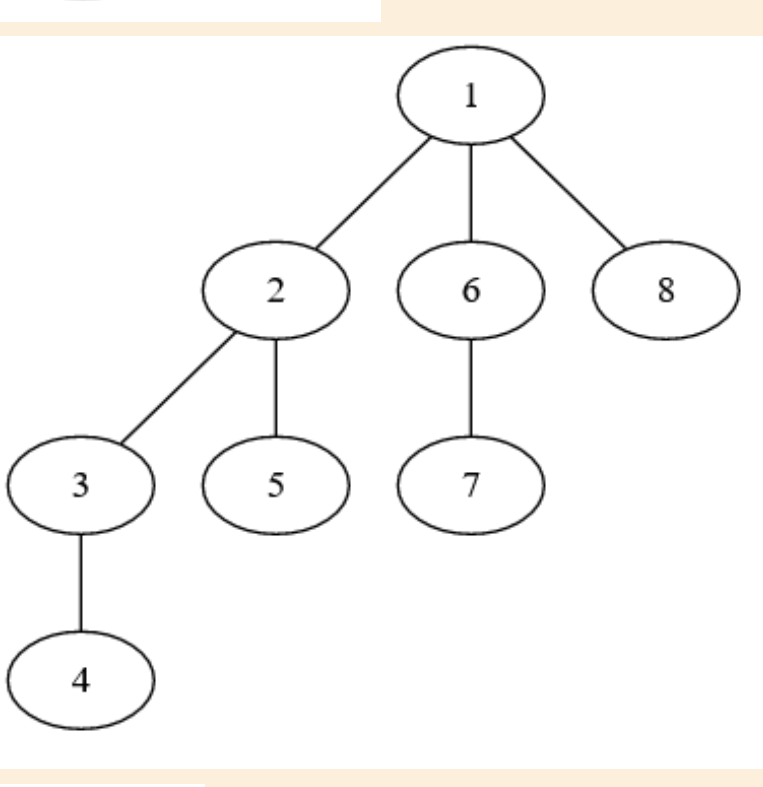
→ Binary heap,



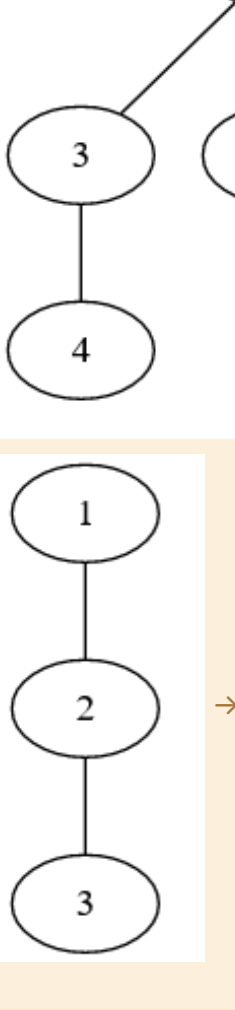
→ Binary heap,



→ Binomial tree of order 2,



→ Binomial tree of order 3,



→ Neither binary heap nor binomial tree

Question **6**
Correct
Mark 1.00 out of 1.00
Flag question

Assuming that we have a binomial heap which contains, among possibly other binomial trees, two binomial trees of order 4. What is the minimum number of nodes in this binomial heap?

Select one or more:

- ☐ 64
- ☐ 16
- ☐ 0
- ☐ 32
- ☐ 4
- ☑ There is no such binomial heap ✓
- ☐ We cannot determine from this information
- ☐ 8

Your answer is correct.

A binomial tree of order 4 has 2^4 nodes (16). So two binomial trees of order 4 have 32 nodes. But in any binomial heap, we can have at most one binomial tree of a given order (two binomial trees of order 4 would become a binomial tree of order 5 in the representation). So there is no such binomial heap.

The correct answer is: There is no such binomial heap

Question **7**
Correct
Mark 1.00 out of 1.00
Flag question

Which of the following values cannot be the number of nodes in a binomial tree?

- ☐ 1
- ☐ 2
- ☑ 3 ✓
- ☐ 4
- ☑ 5 ✓
- ☐ Any positive nonzero number can be the number of nodes in a binomial tree.

Your answer is correct.

The number of nodes in a binomial tree is always a power of 2. So values which are not powers of 2 cannot be numbers of nodes in a binomial tree.

The correct answers are:
3,
5

Question **8**
Correct
Mark 1.00 out of 1.00
Flag question

How many nodes are in a binomial tree of order 5?

Answer: 32 ✓

The number of nodes in a binomial tree of order k is 2^k.

The correct answer is: 32

Question **9**
Correct
Mark 1.00 out of 1.00
Flag question

How many binomial trees are in a binomial heap containing 37 elements?

Answer: 3 ✓

The number of binomial trees is the number of 1s in the binary representation of the number of elements. 37 = 32 + 4 + 1, so there are 3 ones in the binary representation and there will be 3 trees.

The correct answer is: 3

Question **10**
Correct
Mark 1.00 out of 1.00
Flag question

If we have a binomial heap with 84 elements, what is the maximum order of a binomial tree in its representation?

Answer: 6 ✓

84 = 64 + 16 + 4, so there will be a binomial tree with 64 nodes and it is of order 6 (2^6 = 64)

The correct answer is: 6

Question **11**
Correct
Mark 1.00 out of 1.00
Flag question

If we have a binomial heap with 52 elements, what is the minimum order of a binomial tree in its representation?

Answer: 2 ✓

52 = 32 + 16 + 4, so the smallest binomial tree will have 4 nodes, so it is of order 2.

The correct answer is: 2

Question **12**
Incorrect
Mark 0.00 out of 1.00
Flag question

What is the complexity of finding the minimum element in a direct address table? Assume that the table contains m positions and there are n elements added in it.

- ☐ $\Theta(1)$
- ☐ $\Theta(n)$
- ☑ $\Theta(m)$ ⊗
- ☐ O(1)
- ☐ O(n)
- ☑ O(m) ✓
- ☐ None of the other versions

Your answer is incorrect.

Remember, direct address table means that we put the element directly to the position given by the key (so we don't use a hash function). Which means that direct address tables are kind of sorted (since the indexes in an array are sorted). And the minimum element is the first element (from left to right) that you have in the table. This is why at this point we can have best case/worst case/average case.

The correct answer is:
O(m)

Question **13**
Correct
Mark 1.00 out of 1.00
Flag question

In an initially empty hash table with m = 11 and collision resolution by separate chaining (and the division method for the hash function), we insert, in the given order, the following elements: 37, 55, 15, 36, 96, 79, 10 ,74, 87, 92, 35, 56, 77, 52, 32. Which positions from the hash table will be empty (contain empty linked list)?

- ☐ 0
- ☐ 1
- ☐ 2
- ☐ 3
- ☐ 4
- ☑ 5 ✓
- ☑ 6 ✓
- ☑ 7 ✓
- ☐ 8
- ☑ 9 ✓
- ☐ 10
- ☐ 11

Your answer is correct.

The correct answers are:
5,
6,
7,
9

Question **14**
Correct
Mark 1.00 out of 1.00
Flag question

In an initially empty hash table with m = 11 and collision resolution by separate chaining (and the division method for the hash function), we insert, in the given order, the following elements: 37, 55, 15, 36, 96, 79, 10 ,74, 87, 92, 35, 56, 77, 52, 32. On which positions the number of elements is closest to the load factor?

- ☐ 0
- ☑ 1 ✓ we have here 56
- ☐ 2
- ☑ 3 ✓ we have here 36
- ☐ 4
- ☐ 5
- ☐ 6
- ☐ 7
- ☐ 8
- ☐ 9
- ☐ 10
- ☐ 11

Your answer is correct.

Load factor is 15/11 = 1.36. Since the number of elements is an integer value, the closest value is 1. So we are looking for the positions where we have 1 element.

The correct answers are:
1,
3

Question **15**
Incorrect
Mark 0.00 out of 1.00
Flag question

Assume that we have an ADT Bag (without frequencies) represented on an initially empty hash table with m = 11 and collision resolution by separate chaining (and the division method for the hash function). In the Bag we insert, in the given order, the following elements: 37, 55, 15, 36, 96, 79, 10 ,74, 87, 92, 35, 56, 77, 52, 32. What is the last value in the linked list on position 4?

- ○ 36
- ● 92 ⊗
- ○ 15
- ○ 37
- ○ 52

Your answer is incorrect.

On position 4 we have the values: 37, 15 and 92. Since the container is a Bag, where we can have duplicates, when we add a new element we don't have to search to see if it is already there. So, in order to make insert efficient, we always insert at the beginning of the linked list, so the elements will actually be in the reverse order (compared to how they are in the initial sequence): 92, 15, 37

The correct answer is:
37

Question **16**
Correct
Mark 1.00 out of 1.00
Flag question

In a hash table with size m = 10 collision resolution with separate chaining we insert 12 elements. What is the load factor (α) of the table?

- ● 1.2 ✓
- ○ This situation is not possible
- ○ 10
- ○ 12
- ○ 0.833

Your answer is correct.

The load factor is defined as n/m, in our example 12/10 = 1.2

The correct answer is:
1.2

Question **17**
Correct
Mark 1.00 out of 1.00
Flag question

In a hash table with size m = 10 collision resolution with separate chaining we insert 12 elements. How many elements do we have on average in every linked list?

- ● 1.2 ✓
- ○ This situation is not possible
- ○ 10
- ○ 12
- ○ 0.833

Your answer is correct.

This is the load factor, which is defined as n/m, in our example 12/10 = 1.2

The correct answer is:
1.2

Question **18**
Correct
Mark 1.00 out of 1.00
Flag question

We have a hash table with size m = 10 collision resolution with separate chaining where the load factor (α) is 2. What is the maximum number of elements that we can have in one linked list in the table?

- ○ 2
- ○ This situation is not possible
- ○ 10
- ● 20 ✓
- ○ 12

Your answer is correct.

The load factor, which is defined as n/m, so in our case 2 = n/10 => n = 20. We have 20 elements in the hash table. The worst case is that all these 20 elements are in the same linked list.

The correct answer is:
20

Question **19**
Incorrect
Mark -1.00 out of 1.00
Flag question

Would the following definition for a hash function be correct? m represents the size of the hashtable.

h(k) = random integer number in the [0, m-1] interval.

- ● Yes ⊗
- ○ No

Your answer is incorrect.

While this is a fast hash function and it indeed respects the simple uniform hashing assumption(any key has the same probability of hashing to any position, no matter where other keys hashed), in practice it cannot be used. We need a function which for a given key always returns the same value and this does not happen here.

The correct answer is:
No

Question **20**
Incorrect
Mark -1.00 out of 1.00
Flag question

Would the following definition for a hash function be correct? m represents the size of the hashtable.

-- have a dynamic array where you keep the already generated hashvalues in key-value pairs (key is the actual key, value is the value of its hash function)

h(k) = the value stored for k in the dynamic array if k is in the dynamic array

a random integer number in the [0, m-1] interval which is then saved in the dynamic array as well, otherwise

- ● Yes ⊗
- ○ No

Your answer is incorrect.

This setting would solve the previous problem, since it will always return the same hash value for a key. The problem is the time (and maybe space) complexity. Every time you need to generate the value of the hash function for a key, you have to go through the dynamic array to see if the key is already there. And that is not a Theta(1) complexity operation. We use hash tables because the average complexity of the operations is Theta(1). If you need to search in a dynamic array for computing the value of the hash function, that good average complexity no longer holds.

The correct answer is:
No

Question **21**
Incorrect
Mark 0.00 out of 1.00
Flag question

What is the complexity of finding the minimum element in a hash table collision resolution with separate chaining? Assume that the table contains m positions and there are n elements added in it.

- ☐ $\Theta(1)$
- ☑ $\Theta(n)$ ⊗
- ☐ $\Theta(m)$
- ☐ O(1)
- ☑ O(n) ⊗
- ☐ O(m)
- ☐ None of the other versions

Your answer is incorrect.

First of all, finding the minimum (in almost every case) is not a problem with best case/worst case/average case complexity: you need to check each and every element in order to see which is the minimum. So the complexity is going to be Theta.

And checking all the elements means on one hand to check all n elements form the hashtable. This suggests a complexity of Theta(n). But since we don't know where those n elements are exactly (on which positions in the hashtable) we will also need to check all m positions. So the actual complexity is going to be Theta(n+m).

The correct answer is:
None of the other versions

Finish review

◀ Lecture 8 Quiz

Jump to...

Lecture 10 Quiz ▶